

Computer Assignment 1: Blind Experiments with Finite Difference Methods

- As explained in the class, you need to conduct numerical experiments with different finite difference schemes by applying them to a linear advection equation

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = 0$$

with five different Initial Conditions (i.e. solution at time $t=0$) as described below. Take the value of wave speed $a = 1.0$ m/s.

- Discontinuous initial solution as described in Figure 1

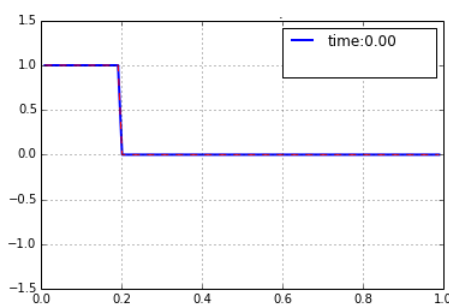


Figure 1: Initial condition 1

$$u(x, 0) = \begin{cases} 1 & \text{if } 0.2 \leq x \\ 0 & \text{Otherwise} \end{cases}$$

- Initial solution having two periods as described in Figure 2

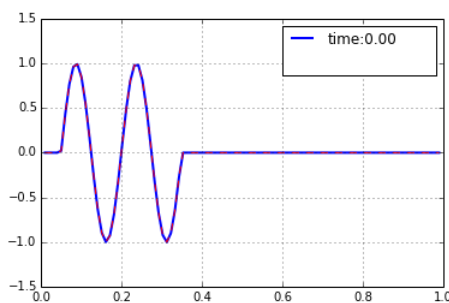


Figure 2: Initial condition 2

$$u(x, 0) = \begin{cases} 0 & \text{for } 0 \leq x < 0.05 \\ \sin(4\pi(\frac{x-0.05}{0.3})) & \text{for } 0.05 \leq x < 0.35 \\ 0 & \text{for } 0.35 \leq x \leq 1 \end{cases}$$

3. Initial solution having four periods as described in Figure 3

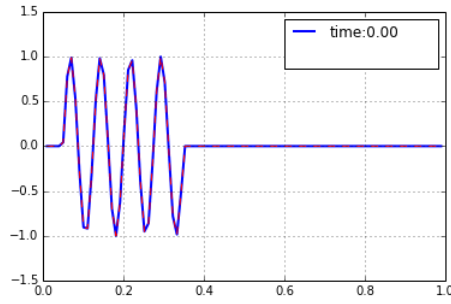


Figure 3: Initial condition 3

$$u(x, 0) = \begin{cases} 0 & \text{for } 0 \leq x < 0.05 \\ \sin(8\pi(\frac{x-0.05}{0.3})) & \text{for } 0.05 \leq x < 0.35 \\ 0 & \text{for } 0.35 \leq x \leq 1 \end{cases}$$

4. Initial solution having six periods as described in Figure 4

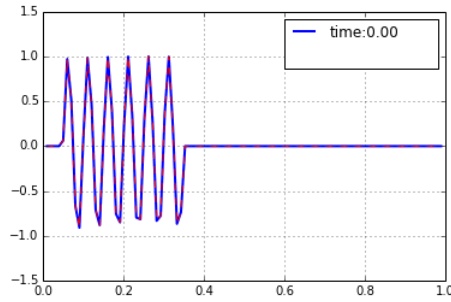


Figure 4: Initial condition 4

$$u(x, 0) = \begin{cases} 0 & \text{for } 0 \leq x < 0.05 \\ \sin(12\pi(\frac{x-0.05}{0.3})) & \text{for } 0.05 \leq x < 0.35 \\ 0 & \text{for } 0.35 \leq x \leq 1 \end{cases}$$

5. Initial solution having Gaussian curve as described in Figure 5

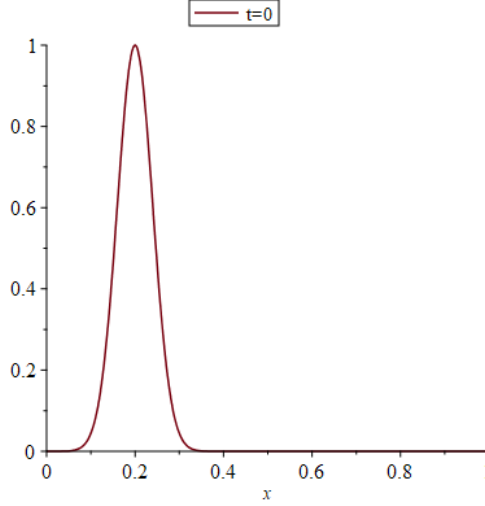


Figure 5: Initial condition 5

$$u(x, 0) = e^{-50 \frac{(x - 0.2)^2}{\sigma^2}} \quad \text{where } \sigma = 0.4$$

- Consider 101 grid points (as shown in Figure ??) for discretizing the computational space $0 \leq x \leq 1$ for all the five initial conditions described earlier.

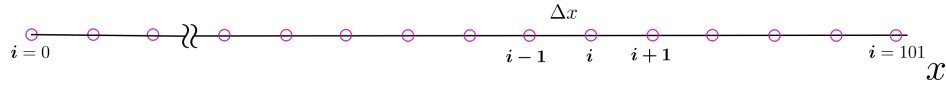


Figure 6: Grid for the linear advection equation.

- Apply the following 7 Finite Difference Schemes (derivations of which will be explained in future) to obtain solutions of the linear advection equations at time $t > 0$ as described below. At present, you need not know the background details of these schemes.

1. Forward Time Forward Space (FTFS) scheme

$$u_i^{n+1} = u_i^n - \nu(u_{i+1}^n - u_i^n)$$

2. Forward Time Central Space (FTCS) scheme

$$u_i^{n+1} = u_i^n - \frac{\nu}{2}(u_{i+1}^n - u_{i-1}^n)$$

3. Forward Time Backward Space (FTBS) scheme

$$u_i^{n+1} = u_i^n - \nu(u_i^n - u_{i-1}^n)$$

4. Lax-Wendroff (LW) scheme

$$u_i^{n+1} = u_i^n - \frac{\nu}{2}(u_{i+1}^n - u_{i-1}^n) + \frac{\nu^2}{2}(u_{i+1}^n - 2u_i^n + u_{i-1}^n)$$

5. Beam-Warming (BW) scheme

$$u_i^{n+1} = u_i^n - \frac{\nu}{2}(3u_i^n - 4u_{i-1}^n + u_{i-2}^n) + \frac{\nu^2}{2}(u_i^n - 2u_{i-1}^n + u_{i-2}^n)$$

6. Fromm (FR) scheme

$$(u_i^{n+1})_{FR} = \frac{1}{2} [(u_i^{n+1})_{LW} + (u_i^{n+1})_{BW}]$$

7. Implicit scheme, known as, Backward Time Central Space (BTCS) scheme

$$-\frac{\nu}{2}u_{i-1}^{n+1} + u_i^{n+1} + \frac{\nu}{2}u_{i+1}^{n+1} = u_i^n$$

where ν is Courant Friedrichs Lewy and it is given by $\nu = \frac{a\Delta t}{\Delta x}$, where Δt is the time step by which solution is advanced. Meaning of ν will be explained later during the course.

- For implementing boundary conditions at the two end of the computational domain (that is, at $x=0$ and $x=1$), set the following:

$$u_0 = 0.0 \quad u_{101} = 0.0$$

for all test cases at all time except for the first one, where the following condition should be applied

$$u_0 = 1 \quad u_{101} = 0.0$$

at all time. However, in case of Beam-Warming, which is not a 3-point scheme, you need to implement boundary conditions in the following manner:

$$u_0 = 0.0 \quad u_1 = 0.0 \quad u_{100} = 0.0 \quad u_{101} = 0.0$$

for all test cases except for the first one, where the following condition should be applied

$$u_0 = 1.0 \quad u_1 = 1.0 \quad u_{100} = 0.0 \quad u_{101} = 0.0$$

at all time.

Since the number of time steps to be considered by you should be such that the disturbances in the solution will not reach the boundaries within that time interval. Consider total time for evolution of solution as $t \approx 0.35$ s.

- By the above statement, it is clear that you need to update your solution (i.e. obtain u_i^{n+1}) only at the interior grid points $i = 2, \dots, 100$.
- While computing solutions, consider three values of ν for each scheme; namely $\nu = 0.5$, $\nu = 1.0$, $\nu = 1.5$.
- Create a report, which should include the following:
 1. The “critical observations” in a tabular form for each results in order to compare different schemes about their behavior for different initial conditions and CFL numbers ν .
 2. All the plots for the solutions obtained at $t \approx 0.35$ s for all the schemes considering all the initial conditions and all the prescribed values of ν .
- Strictly use the plotting range for x-axis as $0 \leq x \leq 1$ and that for y-axis as $-1.5 \leq u \leq 1.5$ (as shown in the Figures 1-5 for initial conditions) uniformly.
- Incorporate all possible explicit finite difference schemes, initial conditions and values of ν in a single code. Similarly, a second code for the implicit scheme. Use Thomas Algorithm for triangular matrix solution arising in the Implicit (BTCS) scheme. You need to submit two codes which will produce results for all the possible cases with user’s option.
- Therefore, your submission folder should contain (1) two computer codes as stated above, (2) a report as explained above and (3) ”README.txt” file which contains the proper description on how to run your codes and get the plots.

General Instructions

- **Checklist for submission:**

- The codes (written in Python/C/C++ only) with proper inline documentation for each function. (Use meaningful variable names).
- You are suggested to use Visual Studio Code (VS Code) IDE for compiling and running your code. This IDE works for any computer language and any operating system.
- A “README.txt” file which contains the proper description on how to run the code and get the plots.
- The plots shown in the figures inside your report must be reproducible independently by the TA’s from your code based on what you have described in “README.txt”.

- **Instruction for submission:**

- Rename your program file as your roll number (example: 184010006.c).
- Keep the Python/C/C++ codes, report and README.txt file (See the checklist) in a folder named it as your roll number (example 184010006a and 184010006b).

- **Notes:**

- There is no grading for this assignment. However, it is compulsory for you to submit the assignment in time.
- Also, your class test on this topic will not be evaluated unless you submit this assignment within due date and your submitted computer programs are working and showing correct results.
- Assignment will not be considered as complete if “instructions for submission” are not followed properly.

- **Copying program from each other will lead to severe penalty.**

- **Warning:** Your computer codes will be checked with plagiarism checker (MOSS software).

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